Project C2

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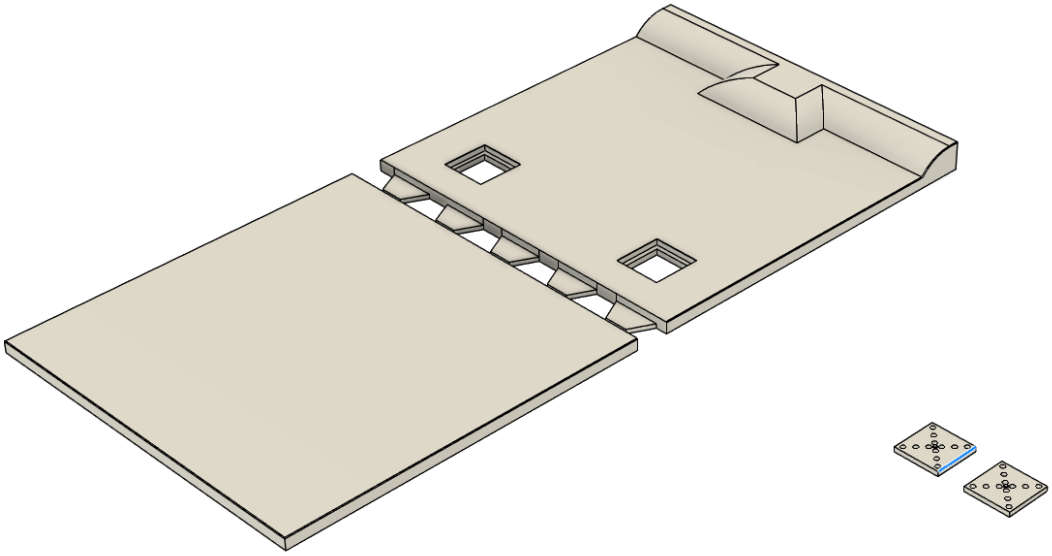


Technical Field:

This project involves the technical fields of additive manufacturing (FDM 3D printing), CAD product design (Autodesk Fusion 360), microclimate / humidity control in sealed enclosures, and fluid behavior in small systems (evaporation, condensation, capillary wicking)

Background Information:

The Capillary Canopy concept came from a very normal hobbyist experience: small tanks are either *open and annoying* (evaporation, dust, constant top-offs) or *covered but dumb* (generic lids that don't manage condensation and don't support a stable internal humidity cycle). The idea is to design and 3D-print (on a Bambu P1P) a set of purpose-built lids for three 2.5-gallon glass tanks, each lid tuned to create a repeatable "mini climate" using geometry—condensation channels, drip rails, optional vents, and a reservoir-fed wick tube—rather than relying on external equipment. All modeling and iteration

happen in Fusion 360, so the whole project lives cleanly in a CAD → prototype → test loop.

Prior Art (Research):

There are barely any options for 2.5-gallon open glass aquariums. There is a solid glass or a metal screen option, basically totally closed or fully ventilated.

My prototype is different because it will provide functionality to the tank instead of just limiting what can enter or exit it. The lid sustains the environment of the tank instead of vice versa.

Project Description:

This project designs and prototypes three 3D-printed lids for three 2.5-gallon glass tanks, modeled in Fusion 360 and printed on a Bambu P1P. Each lid will be a tuned variant (ex: arid / temperate / humid behavior) using the same core architecture: a sealed rim interface, controlled vent options, and a reservoir-fed wick tube that supports passive moisture movement. The lids will incorporate surface features—like internal slopes, drip rails, and condensation channels—to intentionally route condensed water (ideally away from the front viewing pane) and reduce user maintenance. The intended users are hobbyists, students, and “I want a tiny ecosystem, but I also want

my weekends back" people who want a stable micro-environment without external pumps or constant attention.

Innovation Claim:

This project is innovative because it uses 3D-printed lid geometry. Not external equipment to create repeatable, tunable microclimates in small tanks, combining condensation routing, optional vent control, and an integrated reservoir + wick tube into a single modular lid system manufactured on a Bambu P1P from Fusion 360 models.

Usage Scenario:

You set up each 2.5-gallon tank with substrate/scape appropriate to the target humidity profile, install the printed lid, fill the lid reservoir, and let the system passively cycle moisture (evaporation → condensation → guided return). Periodically you top off the reservoir (frequency depends on the variant and ambient room conditions), and you can tweak behavior by swapping vent inserts or switching to a different lid variant.

SIP Brief Part 2

Evaluation Criteria:

- Does the lid fit securely within the 11.0625" × 4.9375" inner lip without excessive play + or - .03.
- Does the lid remain structurally rigid under normal handling without visible flexing?
- Does the lid rest evenly on the black rim without rocking?
- Does the lid remain seated after 30 days without warping?
- Does the system operate without external pumps, misters, or electronics?
- Does the lid prevent insect entry through openings larger than 2 mm?
- Does the lid maintain structural integrity in high-humidity conditions (over 80%)?
- Can the lid be removed and reinstalled without damaging the tank rim?
- Are all components manufacturable using Fusion 360?
- Can wick inserts be removed and replaced without damaging the printed tube?
- Are all internal channels free of stagnant water accumulation after 7 days?
- Does the Arid lid reduce soil moisture retention compared to the Tropical lid?

Goals and Tasks Associated with the Project:

Goal 1: Develop Biome-Specific Passive Lid Designs:

Design three distinct lid geometries (Arid, Temperate, Tropical) that modify internal condensation behavior without altering tank water volume.

Goal 1: Tasks – Biome Lid Design

- Measure tank outer and inner lip dimensions.
- Create parametric base model in Fusion 360.
- Model Arid interior slope geometry.
- Model Temperate dome and perimeter channel.
- Model Tropical ribbing and deep basin.
- Define channel slope angles (1–5°).
- Model return tube placement and diameter.
- Perform section analysis to verify water paths.
- Adjust interior geometries for print feasibility.

Goal 2: Engineer a Secure and Repeatable Fit:

Create a dual-flange lid structure that reliably interfaces with the standard 2.5-gallon Aqueon or other basic open glass black rimed aquarium.

Goal 2: Tasks – Fit Engineering

- Design upper support flange.
- Design lower insert flange.
- Define tolerance (0.015"–0.025").
- Prototype tolerance test print ring.
- Adjust CAD dimensions based on physical test.
- Verify snap-in seating stability.
- Test lid removal and reinsertion durability.

Goal 3: Prototype and Manufacture Using Additive Fabrication

Produce functional physical prototypes using Fusion 360 and a Bambu P1P printer.

Goal 3: Tasks – Fabrication

- Slice models in Bambu Studio.
- Optimize orientation for minimal support.
- Define wall thickness and infill.
- Print Arid prototype.
- Print Temperate prototype.
- Print Tropical prototype.
- Post-process and clean internal channels.
- Install wick inserts (Tropical).

Goal 4: Validate Performance with Controlled Testing

Measure humidity, condensation behavior, and soil moisture stability over a 4-week test period.

Goal 4: Tasks – Testing & Data Collection

- Set up three identical tanks. (Possibly 4, with 2 Tropical; 1 wick vs 2 wick)
- Use identical substrate and water volume.
- Place tanks in same environmental conditions.
- Measure daily humidity.
- Record condensation patterns.
- Observe soil moisture levels.
- Document glass clarity.
- Track wick flow behavior. (Tropical)
- Record 4-week data.
- Compare results between designs.

Goal 5: Demonstrate Environmental Control Through Geometry

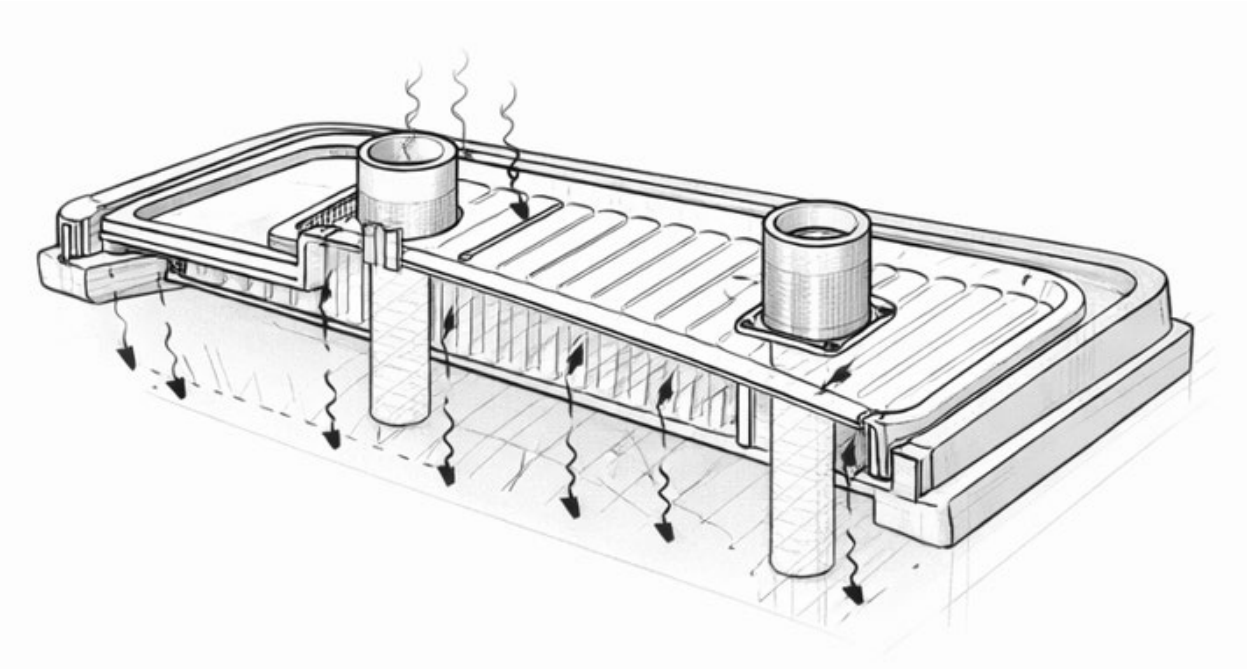
Prove that climate variation can be achieved solely through structural design without mechanical or electrical systems.

Goal 5: Tasks – Validation & Documentation

- Photograph prototypes.
- Create CAD renders.
- Compile humidity comparison table.
- Analyze differences between lid types.
- Record conclusions.
- Prepare presentation slides.
- Delivery of final presentation.

Description of Design Prototype:

Through evaporation, moisture rises through vents in the internal lid components, circulating back into the lid itself, it then pools and returns to the tank through specific channels that guide it.



SIP Brief Part 3 – to be filled out in SIP409

Evaluation Plan:

Project Completion Assessment:

Appendices: